3.] ISLR 6.6.5

Let $\beta = (\beta_0, \beta_1, \beta_2)$ where $\beta_0 = 0$.
And. $x_{11} = x_{12}$, $x_{21} = x_{22}$, $x_{11} + x_{21} = 0$
$y_1 + y_2 = 0$.

3.a] Ridge Optimization
Standard ridge regression minimizes RSS + penalty.

$$\min_{\beta_1, \beta_2} \sum_{i=1}^{2} \left(y_i - \beta_1 x_{i1} - \beta_2 x_{i2}\right)^2 + \lambda\left(\beta_1^2 + \beta_2^2\right).$$

Using $x_{i1} = x_{i2}$, we get the equivalent

$$\min_{\beta_1, \beta_2} \left(y_1 - (\beta_1+\beta_2)x_{11}\right)^2 + \left(y_2 - (\beta_1+\beta_2)x_{21}\right)^2 + \lambda\left(\beta_1^2 + \beta_2^2\right).$$

3.b] Ridge : $\hat{\beta}_1 = \hat{\beta}_2$

→ The residual term depends only on the sum $\beta_1 + \beta_2$.
→ The penalty $\beta_1^2 + \beta_2^2$ is minimized, for any fixed sum, by splitting that sum equally : $\beta_1 = \beta_2$.
→ Hence the unique ridge solution satisfies $\hat{\beta}_1 = \hat{\beta}_2$.

3.c] Lasso Optimization

$$\min_{\beta_1, \beta_2} \sum_{i=1}^{2} (y_i - \beta_1 x_{i1} - \beta_2 x_{i2})^2 + \lambda(|\beta_1| + |\beta_2|)$$

or, using $x_{i1} = x_{i2}$:

$$\min_{\beta_1, \beta_2} (y_1 - (\beta_1 + \beta_2) x_{11})^2 + (y_2 - (\beta_1 + \beta_2) x_{21})^2 + \lambda(|\beta_1| + |\beta_2|)$$

3.d] Lasso : non-unique solutions.

→ The squared error part depends only on $c = \beta_1 + \beta_2$. The optimal c is the same regardless of how it is split between β_1 and β_2.

→ The L_1 penalty $|\beta_1| + |\beta_2|$ also depends only on $|\beta_1| + |\beta_2|$. If β_1 and β_2 share the same sign, then
$$|\beta_1| + |\beta_2| = |\beta_1 + \beta_2| = |c|.$$

→ Therefore any pair (β_1, β_2) satisfying $\beta_1 + \beta_2 = c$, β_1, β_2 have the same sign; yields identical RSS and identical penalty. In particular, if $c > 0$, the entire family
$$\{(\beta_1, \beta_2) : \beta_1 = t, \beta_2 = c - t, \ 0 \le t \le c\}$$
are all equally optimal. (Similarly for $c < 0$.)